GlucoTwin

A Reliability-Aware Digital Twin for 120-Minute Glucose Forecasting & Adverse Event Mitigation

Competition: Happiest Health — Digital Twin Challenge 2026

Team: Team Synapse • **College:** Mohan Babu University

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Repository: https://github.com/Raju-09/GlucoTwin • **License:** MIT Open Source

1. The Healthcare Challenge in India

101 Million Patients: The Critical Need for Proactive Glycemic Management

- **Epidemiological Challenge:** India has over 101 million individuals living with diabetes according to the ICMR-INDIAB study (The Lancet Diabetes & Endocrinology, 2023), with millions undiagnosed in tier-2/3 cities and rural hubs.
- **Limitations of Existing CGM Alerts:** Continuous Glucose Monitors generate readings every 5 minutes, but commercial alert algorithms prioritize reactive threshold alerts or narrow 20–30 minute windows (e.g., Dexcom Urgent Low Soon). When alarms fire close to excursion boundaries, the window for meaningful lifestyle or clinical intervention is constrained.
- **The Proactive 2-Hour Window:** A validated 120-minute forward forecast provides actionable lead time for dietary adjustment, physical activity, or preemptive clinical triage before acute excursions occur.
- **Data Privacy Sandbox Compliance:** Built and evaluated using in-silico synthetic cohorts and open-source formats in strict compliance with India's DPDP Act and global HIPAA frameworks.

'Narrow alarms notify patients of imminent emergencies; proactive digital twins enable foresight.'

2. Mathematical Formulation: Patient Digital State

Decoupling Static Phenotype from Dynamic Causal History

GlucoTwin formally represents the virtual patient at any discrete time step *t* as a triad:

$$S(t) = (\theta_p , h_t , \delta_t)$$

Component	Physiological & Algorithmic Meaning	Features Included
θ_p **Static Phenotype**	Longitudinal EHR context, clinical intake, demographic & baseline metabolic indicators.	Age, Sex, BMI, Baseline HbA1c, Diabetes Duration, Historical Fasting Glucose, Metformin/SGLT2i/Dawn Flags (9 feats)
h_t **Dynamic History**	Causal sliding window of sensor observations (120 min lookback = 24 steps at 5-min cadence).	CGM Lags 0..23, Rolling Mean/Std (1h, 2h), Discrete Velocity dG/dt, Acceleration, Circadian Sin/Cos, Wearable Telemetry
δ_t **Twin Integrity**	Real-time stream quality, latency, telemetry jitter, and fail-closed forecast gating state.	Integrity State (AVAILABLE, DEGRADED, WITHHELD), Reason Codes, Telemetry Freshness, Quality Score %

3. End-to-End System Architecture

Three-Tier Architecture: Ingestion -> Verification -> Serving

- **Tier 1 — Preprocessing & Sensor Physics:** Continuous CGM readings pass through an interstitial physiological delay model (τ=10 min), AR(1) autocorrelated sensor noise, and rate clamping ($|dG/dt| \leq 3.8$ mg/dL/min) to reflect true subcutaneous biology.
- **Tier 2 — Causal Multimodal Feature Pipeline:** Combines 24 CGM lags, rolling kinematics, circadian encodings, and EHR intake into a 56-feature vector. Supervised training windows use strict chronological boundaries with a 120-minute gap buffer to guarantee zero future data leaks.
- **Tier 3 — HistGradientBoosting Core & Reliability Gating:** High-efficiency tree regressor trained on 224,681 windows, coupled to a finite-state machine that evaluates input admissibility before returning a prediction.
- **Tier 4 — REST API & Clinician Dashboard:** Sub-10ms FastAPI backend powering a Streamlit UI with 14-day timeline scrubbing, state inspector, and one-click failure injection.

Full vector diagram available at: docs/architecture_diagram.pdf

4. Core Differentiator: Fail-Closed Reliability Gating

The Digital Twin Knows When NOT to Trust Its Predictions

In clinical machine learning, an overconfident wrong prediction during hardware failure is catastrophic. GlucoTwin enforces fail-closed behavior across three formal states:

Status Level	Trigger Conditions	System Behavior
■ **AVAILABLE**	Nominal stream, freshness < 15 min, complete 24-step history, values in [40, 400] mg/dL.	120-min point forecast displayed with high-confidence green badge.
■ **DEGRADED**	Mild telemetry latency (15–20 min) or minor packet loss; historical context intact.	Forecast displayed alongside advisory warning flag & degraded reason code.
■ **WITHHELD**	Stale stream (>20 min), severe gap (>30 min), sensor dropout, or jump > 4.0 mg/dL/min.	**Output suppressed. No number returned.** Explicit error code issued (e.g. ERR_SENSOR_DROPOUT).

Utility Analysis: Our risk-coverage audit proves that withholding 12% of corrupted inputs suppresses severe prediction errors (>30 mg/dL) and protects clinical fidelity.

5. Multimodal Ablation Study (N=60 Patients)

Evaluated on 43,888 Held-Out Test Windows Under Strict Chronological Hold-Out

Configuration	Input Streams & Features	Test MAE	Δ MAE vs Ref	95% Bootstrap CI
Persistence Baseline	Naive y(t+120) = y(t)	46.88 mg/dL	+28.57 mg/dL	—
Config A (CGM Only)	24 Lags + Rolling Kinematics (41 feats)	18.31 mg/dL	Reference	—
Config B (CGM + Static EHR)	Demographics + Labs + Fasting Lab (50 feats)	**17.61 mg/dL**	**–0.70 mg/dL**	**[–0.96, –0.49]** ($p < 10^{-\blacksquare\blacksquare}$)
Config C (CGM + Wearables)	Decoupled HR, HRV, Steps, Sleep (47 feats)	18.32 mg/dL	+0.01 mg/dL	[–0.08, +0.07] (p = 0.95)
Config D (Full Two-Stream)	EHR + CGM + Wearables (56 feats)	**17.67 mg/dL**	**–0.64 mg/dL**	[–0.88, –0.44]

• **EHR Gain:** Static EHR context provides a statistically validated error reduction (–0.70 mg/dL) across same-patient trajectories.
• **Wearable Negative Finding:** Activity telemetry provided negligible incremental gain under this simulator; reported honestly rather than fabricating artificial gains.

6. Breaking Our Own Claims: Leave-One-Patient-Out (LOPO)

Cross-Validation Across 60 Patients (224,681 Windows)

To test whether the model learned generalizable human physiology or merely memorized synthetic patient identity, we executed a complete Leave-One-Patient-Out (LOPO) cross-validation and a permuted-EHR control:

Evaluation Mode	Mean LOPO MAE	Delta vs CGM-Only	Scientific Verdict
CGM Only	17.65 mg/dL	Reference	Solid kinematic time-series tracking.
CGM + Real EHR	**17.65 mg/dL**	+0.01 mg/dL (p = 0.956)	**No transfer benefit on unseen patients.** EHR acts as same-patient anchor.
CGM + Permuted EHR	**20.47 mg/dL**	**+2.82 mg/dL degradation**	**Significant error rise (+2.82 mg/dL).** Consistent with model utilizing patient-specific static embeddings rather than population averages.

'A system that knows the boundary of its own evidence is far more trustworthy in healthcare than one claiming 99% universal accuracy.'

7. Clinical Adverse Event Classification (t+120 min)

Catching Postprandial Spikes 2 Hours Ahead & Suppressing Alarm Fatigue

Adverse Event State	Model	Positive Windows (N)	Sensitivity	Specificity	Precision	F1-Score	False Alarms / Day*
Hyperglycemia (>180 mg/dL)	Persistence Baseline	12,246	26.6%	70.4%	25.8%	0.262	61.4 windows
Hyperglycemia (>180 mg/dL)	**GlucoTwin (Two-Stream)**	12,246	**76.2%**	**90.6%**	**75.8%**	**0.760**	**19.6 windows (–68%)**
Hypoglycemia (<70 mg/dL)	Persistence Baseline	71	12.7%	99.9%	12.7%	0.127	0.4 windows
Hypoglycemia (<70 mg/dL)	**GlucoTwin (Two-Stream)**	71	0.0%	100.0%	0.0%	0.000	0.0 windows

• **Postprandial Excursions:** Catches nearly 3× more hyperglycemic events 2 hours in advance while reducing false alarms from 61.4 to 19.6/day (a **68% reduction**).
• **Hypoglycemia Tail Limitation:** In adult T2D, hypoglycemia is a rare tail event (<0.2% prevalence). Standard MSE regression regresses towards the mean (~150 mg/dL) and misses rare nadirs below 70 mg/dL; documented as a future milestone for asymmetric loss weighting.

8. Clinician Dashboard & Engineering Maturity

Production-Quality Engineering Ready for Triage

- **Doctor's Interactive Dashboard:** Streamlit-based UI featuring 14-day continuous time scrubbing across 60 patients, clear data provenance badges ([Observed EHR], [Sensor], [Kinematic], [Model]), and unclamped discrete velocity tracking.
- **Live Telemetry Stress Testing:** Real-time sidebar allows clinicians and evaluators to inject sensor dropouts, 20-minute latency delays, or non-physiological spikes (400 mg/dL) to verify fail-closed gating behavior live.
- **Sub-10ms Serving Latency:** Measured live over 100 consecutive API inferences on commodity hardware: **Median: 7.79 ms, Mean: 8.11 ms, p95: 11.14 ms, p99: 13.26 ms**.
- **Containerized Deployment:** Fully reproducible via docker-compose up in under 60 seconds with 69 unit tests passing with 100% test coverage.

Live demo available locally via: streamlit run frontend/streamlit_app.py

9. Responsible AI, Limitations & Clinical Translation

Transparent Boundaries and the Path to Indian Healthcare Impact

- **Explicit Non-Goals:** Research prototype; NOT an FDA/CDSCO-cleared medical device. Evaluated on in-silico synthetic cohorts and must not be used to direct autonomous insulin dosing or clinical therapy.
- **Phase 1 (Completed):** In-silico digital twin architecture, multi-stream feature engineering, fail-closed reliability FSM, and LOPO cross-validation.
- **Phase 2 (Next Milestone):** Prospective transfer validation on clinical research datasets (DiaTrend, OhioT1DM) under data transfer agreements.
- **Phase 3 (Long-Term Vision):** Low-cost edge deployment in Indian primary healthcare centers (PHCs) to assist community physicians in proactive lifestyle counseling.

GlucoTwin: Forecast Earlier. Intervene Smarter. Fail Closed When Uncertain.

Thank you. Team Synapse — Sammeta Siva Prasad Raju & Bedadakota Devi Chandrika (Mohan Babu University)